WORKED SOLUTION

STEP III 1989, Question 6

Pure • Polar Equations of Conics

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q6

The key idea

In polar coordinates a line not through the origin has $1/r$ equal to a combination of $\cos\theta$ and $\sin\theta$, and writing it as $A\cos\theta + B\cos(\theta-\gamma)$ lets us match the conic's own $\cos\theta$ term exactly. Once the $e\cos\theta$ parts agree, the chord equation only has to fix up a single cosine, and symmetry about $\theta = \gamma$ does the rest. Letting the chord shrink to a point gives the tangent, and the tangent equations make the bisector property almost automatic.

Solution

Throughout, (r,θ) are polar coordinates with $x = r\cos\theta$, $y = r\sin\theta$, and we allow $r < 0$ (the point (r,θ) is the same as $(-r,\theta+\pi)$). The conic has its focus S at the origin.

Lines in polar form

Since $lx + my = 1$ cannot pass through the origin, $r \neq 0$ on it, and dividing by r gives

$$\frac{1}{r} = l\cos\theta + m\sin\theta.$$

Expanding the proposed form,

$$A\cos\theta + B\cos(\theta-\gamma) = (A + B\cos\gamma)\cos\theta + B\sin\gamma\,\sin\theta.$$

These agree for all θ if $B\sin\gamma = m$ and $A + B\cos\gamma = l$. Since $\sin\gamma \neq 0$ we may take

$$B = \frac{m}{\sin\gamma}, \qquad A = l - m\cot\gamma,$$

which is the required polar equation. Conversely, any equation $1/r = A\cos\theta + B\cos(\theta-\gamma)$ with A, B not making both coefficients vanish is a line $lx + my = 1$ with $l = A + B\cos\gamma$, $m = B\sin\gamma$. Note that the form is unchanged by $(r,\theta) \to (-r,\theta+\pi)$, so it describes the whole line whichever sign of r we use.

The chord PQ

Look for the chord in the form $a/r = A\cos\theta + B\cos(\theta - \gamma)$. At P and Q we have $\theta = \gamma \mp \delta$ and $a/r = 1 + e\cos\theta$, so we need

$$A\cos(\gamma \mp \delta) + B\cos(\mp\delta) = 1 + e\cos(\gamma \mp \delta).$$

Because $\cos(\mp\delta) = \cos\delta$ for both points, the choice $A = e$, $B\cos\delta = 1$ satisfies both equations at once, and $\cos\delta \neq 0$ lets us solve for B. Two distinct points determine a line, so the chord PQ is

$$\boxed{\frac{a}{r} = e\cos\theta + \sec\delta\,\cos(\theta - \gamma).}$$

(If $\cos\delta = 0$ then P and Q are at $\theta = \gamma \mp \frac{\pi}{2}$, and PQ is a focal chord; a line through the origin has no equation of the form $1/r = \ldots$, which is why that case is excluded.)

Ask yourself How do we know the chord meets the conic only at P and Q?

At each θ the conic has exactly one point, with $a/r = 1 + e\cos\theta$. The chord passes through it exactly when $1 + e\cos\theta = e\cos\theta + \sec\delta\cos(\theta - \gamma)$, that is, when $\cos(\theta - \gamma) = \cos\delta$, so $\theta = \gamma \pm \delta$. The $e\cos\theta$ terms cancel, which is the whole point of choosing $A = e$.

The tangent at $\theta = \gamma$

The tangent at the point G where $\theta = \gamma$ is the limit of the chord PQ as P and Q close in on G, that is, as $\delta \to 0$. The chord equation depends continuously on δ and $\sec\delta \to 1$, so the tangent is

$$\boxed{\frac{a}{r} = e\cos\theta + \cos(\theta - \gamma).}$$

As a confirmation, this meets the conic where $\cos(\theta - \gamma) = 1$, i.e. only at G, as a tangent should. More decisively, in cartesian form the tangent is $(e + \cos\gamma)x + (\sin\gamma)y = a$, with normal vector $(e + \cos\gamma,\ \sin\gamma)$. The conic is $r = a - ex$; differentiating along the curve gives $(x\,\mathrm{d}x + y\,\mathrm{d}y)/r = -e\,\mathrm{d}x$, i.e. $(e + x/r)\,\mathrm{d}x + (y/r)\,\mathrm{d}y = 0$. At G, $x/r = \cos\gamma$ and $y/r = \sin\gamma$, so the curve's normal there is the same vector.

The final part

Put the focus S at the origin, so the conic is $a/r = 1 + e\cos\theta$, and let L and M be the points with $\theta = \alpha$ and $\theta = \beta$. Write

$$\gamma = \tfrac{1}{2}(\alpha + \beta), \qquad \delta = \tfrac{1}{2}(\beta - \alpha),$$

so that L and M are at $\theta = \gamma - \delta$ and $\gamma + \delta$, exactly as for P and Q above. By the previous result the tangents at L and M are

$$\frac{a}{r} = e\cos\theta + \cos(\theta - \alpha), \qquad \frac{a}{r} = e\cos\theta + \cos(\theta - \beta).$$

At T both hold, so $\cos(\theta-\alpha)=\cos(\theta-\beta)$. Hence either $\theta-\alpha=\theta-\beta+2k\pi$, which would make $L=M$, or

$$\theta-\alpha=-(\theta-\beta)+2k\pi \quad\Longrightarrow\quad \theta=\tfrac{1}{2}(\alpha+\beta)+k\pi=\gamma+k\pi.$$

So T lies on the line through S in the direction $\theta=\gamma$, with

$$r_T=\frac{a}{e\cos\gamma+\cos\delta}$$

(the tangents are parallel, and T does not exist, exactly when $e\cos\gamma+\cos\delta=0$).

The rays SL and SM point in the directions $\alpha=\gamma-\delta$ and $\beta=\gamma+\delta$ (taking the usual case $r>0$ at L and M, as for an ellipse or parabola). The line $\theta=\gamma$ makes the angle δ with each of them, so the line ST bisects the angle LSM. (If $r_T<0$, T lies on the opposite ray, and ST bisects the reflex angle LSM; the angles LST and MST are still equal.)

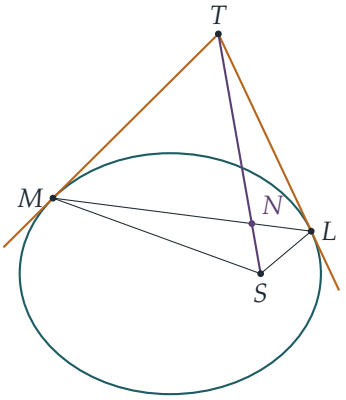
To find where ST meets LM, call the point N. The chord LM is $a/r=e\cos\theta+\sec\delta\cos(\theta-\gamma)$, and on ST we have $\theta=\gamma$, so

$$\frac{a}{SN}=e\cos\gamma+\sec\delta \quad\Longrightarrow\quad SN=\frac{a\cos\delta}{1+e\cos\gamma\cos\delta}.$$

Using $2\cos\gamma\cos\delta=\cos(\gamma-\delta)+\cos(\gamma+\delta)=\cos\alpha+\cos\beta$, this is

$$\boxed{N \text{ is at } \theta=\tfrac{1}{2}(\alpha+\beta),\quad r=\frac{2a\cos\frac{1}{2}(\beta-\alpha)}{2+e\cos\alpha+e\cos\beta}.}$$

When LM is a focal chord ($\cos\delta=0$) this gives $N=S$, as it must.



Checks

- Try a circle ($e=0$, radius a): the tangent becomes $a/r=\cos(\theta-\gamma)$, i.e. $x\cos\gamma+y\sin\gamma=a$, the familiar tangent to $x^2+y^2=a^2$.

- The angle bisector theorem in triangle LSM says $LN : NM = SL : SM$. With $a = 1.5$, $e = 0.6$, $\alpha = 40^\circ$, $\beta = 160^\circ$ (the figure), $SN \approx 0.791$, and the ratio checks numerically.
- A computer tested the chord and tangent equations, the direction of ST, the angle equality and the formula for SN on many thousands of random conics and points.

Common slips

- Solving $\cos(\theta - \alpha) = \cos(\theta - \beta)$ with only the $+$ sign. That branch gives nothing (it needs $\alpha = \beta$), and the useful solution is the $-$ sign.
- Not using the freedom in A: taking $A = e$ is what makes the $e\cos\theta$ terms cancel when intersecting with the conic.

Beyond STEP

The result that ST bisects $\angle LSM$ is a classical focal property of conics, and the line LM is the *polar* of T with respect to the conic. Pole–polar duality is the gateway to projective geometry, where all non-degenerate conics are equivalent and tangency and incidence become purely algebraic. Brannan, Esplen and Gray's *Geometry* is a good first-year account of conics and projective geometry.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.